SeizeLUT: A LUT-Native Multimodal Seizure Detector for Wearable Devices

B.Tech Major Project, Phase 1 Report

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Abstract

Automatic seizure detection with wearable sensors could give people with epilepsy objective seizure counts and timely alarms, but the detector must run for days on a small battery next to the sensors. This project asks whether a **LUT-native** neural network, whose neurons are the lookup tables (LUTs) that make up an FPGA, can detect seizures from behind-the-ear EEG, ECG, EMG and motion signals in patients it has never seen, with no multipliers in the classifier and a budget of a few thousand LUTs.

We use SeizeIT2, a public wearable dataset of 125 patients with focal epilepsy (11,626 hours, 883 seizures), with its official split into 96 training and 29 test patients. Phase 1 covered: a complete audit of the dataset, which found that it differs from its documentation (256 Hz instead of 250 Hz, 883 instead of 886 seizures); an independent reproduction of the published ChronoNet baseline, which was only partly reproduced and whose released code selects its model on the test patients; a bit-exact integer feature pipeline of 117 features that loses no accuracy against floating point (AUROC 0.736 vs 0.735); and a comparison of nine model families with patient-wise cross-validation and event-based SzCORE metrics.

So far, on cross-validation of the 96 training patients, a Differentiable Weightless Neural Network of 50 six-input LUTs reaches an event-F1 of 0.211 with 2.4 false alarms per day at an estimated 960 LUTs and no multipliers. It is seven to eight times smaller than the next LUT-based models (TreeLUT, logic-gate network) and better than an INT8 neural network that needs 9,600 multiplications per decision, but below gradient-boosted trees (event-F1 0.292). It detects 88 % of tonic-clonic seizures but few focal seizures, and depends on the ECG signal. An interpretability study of the network led to a better feature selection that raised its event-F1 from 0.148 to 0.208.

The literature survey of 79 verified references found no LUT-native seizure detector and no LUT network that fuses modalities or handles missing sensors. Phase 2 will complete the extensions of the network, evaluate the final models once on the 29 held-out test patients, and implement the detector on an FPGA.

Keywords: epilepsy, seizure detection, wearable EEG, behind-the-ear EEG, multimodal fusion, weightless neural networks, lookup tables, FPGA, SeizeIT2.

Chapter 1. Introduction

1.1 Background

Epilepsy is a common neurological disorder: an estimated 45.9 million people had active epilepsy in 2016 [1]. Seizures are classified by onset; focal seizures start in one part of the brain, may or may not impair awareness, and can spread into bilateral tonic-clonic seizures [2]. Patients are often unaware of their own seizures, so seizure diaries undercount them [3], and tonic-clonic seizures carry a risk of sudden unexpected death in epilepsy [4]. Reliable, automatic seizure detection outside the hospital would give clinicians objective seizure counts and caregivers timely alarms.

In hospital, seizures are diagnosed with video-EEG from a full scalp electrode cap, which is not usable at home for days. Wearable devices are recommended today only for detecting tonic-clonic seizures [5]; for focal seizures that do not become tonic-clonic (822 of the 883 seizures in SeizeIT2) the problem is open, and false alarms are the main obstacle [6]. Behind-the-ear EEG combined with ECG, EMG and motion sensors is a promising, discreet alternative [7], [8], and the SeizeIT2 dataset [9] now makes it possible to develop and evaluate detectors on 125 patients recorded with such a wearable.

1.2 Why a dedicated hardware detector

A wearable seizure detector has to run for days on a small battery, close to the sensors, and raise an alarm within seconds. Sending raw signals to a phone or a server costs radio energy and depends on a connection; running a neural network on a microcontroller or an accelerator costs multiplications, which dominate the area and power of digital hardware. Field-programmable gate arrays (FPGAs) offer a different trade-off: their basic logic element is a small lookup table (LUT), typically with six inputs, that can implement any Boolean function of those inputs in one clock cycle.

A family of recent machine-learning models, which we call **LUT-native**, is built from exactly this element: weightless neural networks, whose neurons are truth tables [10], logic-gate networks [11], and LUT-based networks designed for FPGAs [12]. They need no multipliers at

inference time, so a classifier can cost a few hundred to a few thousand LUTs. They have been applied to image benchmarks, physics triggers and, very recently, ECG [13] and EEG [14] classification, but, to our knowledge, not to seizure detection on multimodal wearable data.

1.3 Problem statement

Can a LUT-native neural network, fed by integer features computed from behind-the-ear EEG, ECG, EMG and motion, detect seizures in patients it has never seen as well as the published SeizeIT2 baselines, with no multipliers in the classifier and a budget of a few thousand FPGA LUTs?

1.4 Objectives

1. Audit the SeizeIT2 dataset [9] completely and reproduce its published baselines.
2. Design an integer-only feature pipeline that a hardware front end can compute bit-exactly, and show that it loses no accuracy against floating point.
3. Compare LUT-native models (DWN, logic-gate networks, TreeLUT, LogicNets-style networks) with software references (gradient-boosted trees) and multiplier-based networks (INT8 MLP, CNN/TCN, GRU) on the same features, splits and metrics, reporting accuracy against estimated hardware cost.
4. Extend the best LUT-native model with ideas specific to wearable seizure detection (memory across seconds, cheaper input encoders, adaptation to new patients, learning from a stronger model, losses for rare and noisily labelled events, robustness to missing sensors, interpretability), keeping only what is statistically significant.
5. Evaluate the final models once on the 29 held-out test patients with event-based metrics (SzCORE [15]).
6. Deliver a hardware handoff package (truth tables, thresholds, test vectors, verified reference Verilog) and implement the detector on an FPGA (Phase 2).

1.5 Expected contributions

- The first differentiable LUT-native seizure detector, and the first benchmark of LUT-family models on SeizeIT2.
- A bit-exact integer multimodal feature front end with its fixed-point specification.
- An independent reproduction of the SeizeIT2 ChronoNet baseline, including a protocol that uses no test information.
- Tier-2 extensions of the DWN for biosignal events (stateful LUT networks, comparator-budgeted encoders, missing-sensor robustness), each judged by per-patient statistics.
- A hardware handoff package that lets the FPGA implementation be checked bit for bit against the software.

1.6 Organisation of this report

Chapter 2 surveys the literature. Chapter 3 describes the dataset and our audit of it. Chapter 4 presents the proposed system and the evaluation protocol. Chapter 5 reports the work done and the results so far, Chapter 6 the challenges we met and how we solved them, and Chapter 7 the plan for Phase 2.

Chapter 2. Literature survey

The survey covers four areas, from the clinical problem to the hardware: (2.1) epilepsy and why seizures need to be detected automatically, (2.2) wearable and multimodal seizure detection, including behind-the-ear EEG and the SeizeIT project, (2.3) detection algorithms and how they are evaluated, (2.4) LUT-native, weightless and logic networks, and (2.5) hardware seizure detectors and efficient inference. Section 2.6 summarises the research gaps and how this project addresses them. Every reference was checked against a published record (title, authors, venue, year); numbers quoted from a paper are as reported in its abstract unless stated otherwise.

2.1 Epilepsy and the need for automatic seizure detection

Epilepsy affected an estimated 45.9 million people with active epilepsy in 2016 [1]. The ILAE 2017 operational classification [2] divides seizures by onset into focal and generalised; focal seizures are further described by whether the patient stays aware (focal aware, focal impaired awareness), and a focal seizure can spread into a focal to bilateral tonic-clonic seizure. These are the classes annotated in SeizeIT2.

Patients are often unaware of their own seizures, which Elger and Hoppe identify as the main cause of seizure under-reporting [3]; seizure diaries therefore undercount, and treatment decisions are made on incomplete information. Seizures are also dangerous: in the MORTEMUS study of sudden unexpected death in epilepsy (SUDEP) in monitoring units, the fatal cardiorespiratory dysfunction followed a generalised tonic-clonic seizure [4]. Objective, automatic seizure counting outside the hospital is therefore a clinical need, both to alert caregivers and to give clinicians reliable seizure counts.

The ILAE–IFCN clinical practice guideline [5] gives a (weak) recommendation for wearable devices that detect generalised and focal to bilateral tonic-clonic seizures, and does not recommend current devices for other seizure types. Detecting **focal** seizures, the majority in focal epilepsy, with a wearable remains an open problem.

2.2 Wearable and multimodal seizure detection

Reviews and devices. A review of multimodal wearable detection [16] found sensitivities from 4 % to 100 % across studies, with tonic-clonic seizures detected best. A wearable EMG device detected tonic-clonic seizures with more than 90 % sensitivity in a 71-patient prospective study [17], and wrist-worn electrodermal-activity and accelerometer sensors reached 94.55 % sensitivity at 0.2 false alarms per day for convulsive seizures [18]. A meta-analysis [19] pooled a sensitivity of 0.91 for tonic-clonic seizures at 2.1 false alarms per 24 hours. All of this evidence concerns convulsive seizures.

Behind-the-ear EEG and the SeizeIT project. EEG is the most direct signal of a seizure, but a full scalp EEG is not wearable for days. The SeizeIT project records EEG from electrodes behind the ears. Vandecasteele et al. [7] showed that seizures can be annotated and detected on behind-the-ear EEG, with a patient-specific detector at 69.1 % sensitivity and 0.49 false positives per 24 h and a patient-independent one at 64.1 % and 2.8 per 24 h. Adding ECG to a two-channel EEG detector raised sensitivity by 8–11 % at the same false-alarm rate [8], and helped human reviewers too: with ECG, false detections fell from 6.5 to 2.4 per day [20]. The ICASSP 2023 Grand Challenge on wearable-EEG seizure detection [21] was won by an ensemble of decision trees. SeizeIT2 [9] then released 125 patients with more than 11,000 hours of behind-the-ear EEG, ECG, EMG and motion, with ChronoNet and SVM baselines; it is the dataset of this project. A multicentre clinical validation of the same wearable on 192 adults [6] reached an algorithmic sensitivity of 0.73 at a precision of only 0.004 (most alarms false) before human review, which shows that **false alarms are the main obstacle** for focal seizures.

Other work on SeizeIT2. A study of tonic-clonic seizures in the SeizeIT2 cohort [22] fused EEG, EMG and accelerometry and reached 90.9 % sensitivity at 0.1 false positives per 24 h, against 100 % at 10.3 per 24 h for EEG alone: fusion mainly removes false alarms. ECG-only anomaly detection [23] and a complex-valued multimodal network (CMEpiNet) [24] have also been applied to SeizeIT2; none of these reports a hardware budget.

Fusion and missing modalities. BrainFuseNet [25] fused EEG, PPG and accelerometry and ran on a microcontroller (64.28 % sensitivity at 0.21 false positives per hour). A cyclic transformer

trained on EEG and EMG but tested on EEG alone [26] is the only work we found that targets a missing modality at test time.

Work	Year	Signals	Seizures / patients	Result (as reported)	Gap for our problem
Beniczky et al. [17]	2018	EMG	tonic-clonic, 71 patients	> 90 % sensitivity	convulsive only
Onorati et al. [18]	2017	EDA + ACC	convulsive	94.55 %, 0.2 FA/day	convulsive only
Naganur et al. [19]	2022	meta-analysis	tonic-clonic	0.91 sens., 2.1 FA/24 h	focal not solved
Vandecasteele et al. [7]	2020	bte-EEG	focal	64.1 %, 2.8 FP/24 h (patient-independent)	EEG only
Vandecasteele et al. [8]	2021	bte-EEG + ECG	focal	+8–11 % sensitivity with ECG	software only
Bhagubai et al. [21]	2024	bte-EEG	challenge	tree ensemble won	no hardware budget
Zhang et al. [22]	2024	EEG + EMG + ACC	tonic-clonic	90.9 %, 0.1 FP/24 h	tonic-clonic only
Swinnen et al. [6]	2026	wearable (SeizeIT2 device)	focal, 192 adults	0.73 sens. at 0.004 precision	false alarms
BrainFuseNet [25]	2024	EEG + PPG + ACC	—	64.28 %, 0.21 FP/h, MCU	multipliers, no missing-sensor test
Zhang et al. [26]	2024	EEG + EMG	—	—	transformer, large model

Gap. Wearable detection is clinically established only for tonic-clonic seizures. For focal seizures, fusion of EEG with ECG, EMG and motion is the most consistent way to raise sensitivity and cut false alarms, but the published fusion detectors are software models or multiply-accumulate networks on microcontrollers, and only one handles a missing sensor.

2.3 Seizure detection algorithms and their evaluation

Feature-based detectors. Shoeb and Guttag [27] trained a patient-specific support vector machine on EEG spectral features and detected 96 % of 173 test seizures on the CHB-MIT scalp EEG database, with a median latency of 3 s and 2 false detections per 24 h. Patient-specific training needs seizures of the same patient, which a wearable for new patients does not have. On wearable behind-the-ear EEG, the winner of the ICASSP 2023 Grand Challenge, SeizFt [28],

used hand-crafted features, Fourier-transform-surrogate augmentation and a gradient-boosted tree ensemble (CatBoost); a random forest on band-power features reached 73.6 % sensitivity and 96.7 % specificity in the same challenge [29]. Tree ensembles are the strongest published approach on SeizeIT data, which is why LightGBM, XGBoost and CatBoost are our software references.

Deep-learning detectors. ChronoNet [30] stacks multi-scale one-dimensional convolutions and densely connected GRU layers on raw EEG; it was designed for abnormal-EEG classification and is the deep baseline of SeizeIT2. In the ICASSP 2023 challenge, ChronoNet trained on SeizeIT1 reached only 15.2 % sensitivity on the SeizeIT2 validation patients [29]. EEGNet [31] is a compact convolutional network for EEG with depthwise and separable convolutions; EEGWaveNet [32] extracts multiscale features with trainable depthwise convolutions and improves subject-independent detection on CHB-MIT and TUSZ; graph neural networks with self-supervised pre-training [33] reached a detection AUROC of 0.875 on TUSZ but need a full electrode montage to build the graph; U-Net transformers [34] and multi-signal transformers such as BIOT [35] push accuracy further with much larger models. These networks are floating-point and multiplier-based; their size is reported in parameters, not in hardware cost.

Learning from imperfect labels. Seizure onset and offset are uncertain even for experts. Weak supervision with noisy clinical labels [36], attention-based multiple-instance learning when only time periods are labelled [37], and uncertainty-aware training with noisy labels (BUNDL) [38] address this on scalp EEG. Knowledge distillation from a full-montage teacher into a two-channel student [39] keeps detection competitive with very few channels. These ideas motivate our extensions N4 (distillation) and N5 (noise-robust and multiple-instance losses), applied for the first time to a LUT network.

Datasets and evaluation. Most detectors are developed on scalp EEG: CHB-MIT and the TUH Seizure Corpus (315 subjects in its 2018 release) [40]. How a detector is scored changes its numbers: epoch-based scoring, any-overlap event scoring (OVLP) and time-aligned event scoring (TAES) give different results for the same system [41]. SzCORE [15] standardised data formats, cross-validation and event-based metrics (sensitivity, precision, F1, false positives per day, any-overlap matching) with an open-source scorer. In the first blind SzCORE challenge [42], on 65 unseen subjects and 4,360 hours of EEG, the best algorithm reached an event-F1 of

43 % (sensitivity 37 %, precision 45 %), and reported results did not hold up on blind data. Data leakage, letting a subject's data appear in both training and test sets, is common in EEG deep learning and inflates results [43].

Work	Year	Method	Data	Result (as reported)	Gap for our problem
Shoeb & Guttag [27]	2010	patient-specific SVM	CHB-MIT	96 % of seizures, 2 FD/24 h	patient-specific, scalp EEG
SeizFt [28]	2023	features + CatBoost	SeizeIT1/2	ICASSP 2023 winner	EEG only, no hardware cost
Huang et al. [29]	2023	random forest; ChronoNet	SeizeIT1/2	RF 73.6 % sens.; ChronoNet 15.2 %	EEG only
ChronoNet [30]	2019	CNN + GRU	TUH abnormal EEG	new benchmark at the time	large, multiplier-based
EEGWaveNet [32]	2022	multiscale CNN	CHB-MIT, TUSZ	better subject-independent	scalp EEG, float CNN
Tang et al. [33]	2022	self-supervised GNN	TUSZ	AUROC 0.875	needs full montage
BUNDL [38]	2026	noisy-label training	TUH, CHB-MIT	more robust to label noise	scalp EEG
Zheng et al. [39]	2023	distilled GNN	TUSZ	competitive with 2 channels	float GNN student
SzCORE [15]	2024	evaluation framework	scalp EEG	standard event metrics	not for wearables
SzCORE challenge [42]	2025	blind benchmark	65 subjects	best event-F1 43 %	scalp EEG

Gap. Strong detectors are mostly patient-specific or scalp-EEG based, floating-point and large, and they are scored in ways that cannot be compared. On blind, unseen patients even the best reach an event-F1 below 50 %. A small detector for wearable multimodal data, scored with SzCORE-style event metrics on strictly held-out patients, is still missing.

2.4 LUT-native, weightless and logic networks

Weightless neural networks. A weightless network replaces weighted sums with memory lookups. WiSARD [44] split a binary input into small tuples, each addressing a RAM node, with one discriminator per class whose score is the number of RAM nodes that fire. Such networks are trained by writing tables, not by gradient descent, and their inputs must already be binary. BTHOWeN [45] added Bloom filters, hashing and a Gaussian thermometer encoding of

continuous inputs, with an FPGA accelerator; ULEEN [46] borrowed training techniques from binarised networks and reported 3.6 times the area-normalised throughput and 7.1 times the energy efficiency of a FINN binarised network on FPGA (abstract).

Differentiable Weightless Neural Networks (DWN) [10] made the lookup tables themselves trainable end to end: thermometer-encoded inputs, LUT layers trained with an Extended Finite Difference gradient, and a learnable input mapping that decides which bits each LUT reads. The authors report better latency, throughput, energy and area than earlier FPGA designs, and better accuracy than XGBoost on a microcontroller under tight memory limits. DWNs have since been applied to wearable inertial human-activity recognition [47], reaching 96.3–96.7 % accuracy at 56–104 nJ per sample (abstract), and to continuous control with thermometer-encoded sensor inputs compiled to FPGA logic [48]. The DWN is the base model of this project.

Logic-gate networks. Deep Differentiable Logic Gate Networks (LGN) [11] learn, for every neuron, which of the 16 two-input Boolean functions to compute, and after training run as pure logic (over a million MNIST images per second on one CPU core, abstract). Convolutional LGNs [49] reach about 86 % on CIFAR-10 with tens of millions of gates; Light LGNs [50] reduce the parameters of higher-fan-in gates; a recurrent LGN [51] is the only stateful logic network we found, evaluated on machine translation (5.00 BLEU against 5.41 for a GRU on WMT’14 English–German, abstract).

LUT-based networks for FPGAs. LUTNet [52] replaced the XNOR operations of binarised networks with trainable K-input LUTs, roughly doubling area efficiency. LogicNets [12] trains sparse networks with few-bit activations so that every neuron can be enumerated as a truth table and synthesised directly into FPGA logic, with latencies below 15 ns; its cost grows exponentially with fan-in, which keeps fan-in tiny. PolyLUT [53] lets each LUT neuron compute a polynomial of its inputs, and NeuraLUT [54] hides a small dense sub-network inside each LUT (up to 4.3 times lower latency at equal accuracy, abstract). AmigoLUT [55] scales LUT networks through ensembles, and ReducedLUT [56] compresses the tables by treating input patterns never seen in training as “don’t care” (up to 1.63 times fewer LUTs, abstract), an assumption that is risky when a new patient produces patterns the training patients did not. TreeLUT [57] maps quantised gradient-boosted trees into LUT-only logic with 4–101 times lower area-delay product than recent LUT networks (abstract); we use it as a strong non-neural LUT baseline.

The cost of the encoder. Mecik and Kumm [58] built a complete DWN hardware generator and found that the thermometer encoder can raise LUT usage by up to 3.2 times and dominates the cost of small networks. Our own cost estimate agrees: in our 50-LUT base model the encoder accounts for about 850 of about 960 LUTs (Chapter 4), which is why we treat encoder design (N2) as a first-class question.

LUT and logic networks on biosignals. Three recent works come closest to ours. Mommen et al. [13] classified ECG arrhythmias inter-patient with LGNs and LUT networks, up to 94.3 % accuracy on MIT-BIH with 2,000–2,990 LUTs on an Artix-7 FPGA (about 5–7 mW, estimated). Dharia et al. [14] applied LGNs to EEG classification (dementia and emotion) on an embedded CPU. The only weightless network applied to seizures is a WiSARD model for seizure anticipation evaluated on 4 patients with accuracy as the only metric [59].

Work	Year	Model	Domain	Hardware / result (as reported)	Gap for our problem
WiSARD [44]	1984	RAM-node weightless net	images	—	no gradient training, binary inputs
BTHOWeN [45]	2022	weightless + Bloom filters	tabular, MNIST	FPGA accelerator	counting-based training, static
ULEEN [46]	2023	weightless	MLPerf Tiny	7.1× energy efficiency vs FINN	no biosignals
DWN [10]	2024	differentiable LUT6 network	tabular	FPGA, MCU	stateless; no biosignal events
nanoML HAR [47]	2025	DWN	wearable inertial HAR	56–104 nJ / sample	single modality, no rare events
LGN [11]	2022	2-input logic gates	images	>1 M images/s on CPU	random connections, no biosignals
Recurrent LGN [51]	2025	stateful logic gates	translation	—	not biosignals, no hardware
LogicNets [12]	2020	sparse quantised LUT net	jet tagging, NID	< 15 ns latency	tiny fan-in, static
NeuraLUT [54]	2024	dense sub-nets in LUTs	jet tagging, MNIST	4.3× lower latency	static
TreeLUT [57]	2025	quantised GBDT in LUTs	tabular, MNIST	4–101× lower area-delay	static per-window
Mecik & Kumm [58]	2025	DWN hardware generator	jet tagging	encoder up to 3.2× LUTs	—
Mommen et al. [13]	2026	LGN, LUT network	ECG arrhythmia	2.0–3.0k LUTs, 5–7 mW (est.)	single modality, beat-level

Work	Year	Model	Domain	Hardware / result (as reported)	Gap for our problem
Dharia et al. [14]	2026	LGN	EEG (dementia, emotion)	embedded CPU	no FPGA, no seizures
de Aguiar et al. [59]	2015	WiSARD	seizure anticipation, 4 patients	—	accuracy only, tiny cohort

Gap. We found no differentiable LUT-native or logic-gate seizure detector. Every LUT, weightless or logic network we found is single-modality and stateless (except one recurrent LGN for language), none handles a missing sensor, and only one [13] tests generalisation to unseen patients. The encoder's hardware cost is rarely reported, although it dominates small networks.

2.5 Hardware seizure detectors and efficient inference

ASICs. The first micro-power seizure-detection chips put the analogue front end and feature extraction on chip: Verma et al. [60] extracted spectral EEG features at 9 µJ per feature vector and sent them to a central classifier, which cut system power 14 times by avoiding raw-EEG transmission. Later systems-on-chip ran the classifier too: a 16-channel patient-specific SVM chip at 2.73 µJ per classification [61], a non-linear SVM chip at 1.83 µJ per classification [62], and an SVM processor with on-chip learning at 1.9 mW [63]. Tree-based classifiers proved particularly cheap: gradient-boosted trees on intracranial EEG at 41.2 nJ per classification [64], and the NeuralTree chip at 0.227 µJ per classification across 256 channels [65]. Trees only compare and add, which is also why they map well to lookup tables.

FPGAs. FPGA detectors include an STFT + SVM design on a Zynq-7000 at 380 mW of programmable-logic power [66], SVM training and classification on a Virtex-7 [67], a patient-specific iEEG detector on a Cyclone V at 0.49 mW dynamic power [68], an ultra-low-complexity line-length detector [69], and a spiking neural network on a small Lattice iCE40UP5K [70], which reported 4.55 nJ and 0.5 µs per inference with 0.3 false positives per hour on CHB-MIT using four temporal EEG channels. The last is the closest to our Phase 2 target: a low-cost FPGA and few channels near the ear.

Microcontrollers. e-Glass [71] put four EEG electrodes in a glasses frame and ran detection on an embedded processor for 2.71 days per charge. EEGformer [72] ran a compact transformer on temporal EEG channels on GAP9 at 0.31 mJ per inference and detected 73 % of CHB-MIT

seizures at 0.15 false positives per hour. BrainFuseNet [25] fused EEG, PPG and accelerometry on the same microcontroller at 0.11 mJ per inference. Tree-based EEG artefact detection on a low-power processor [73] and the BioWolf ExG platform [74] show the same push towards wearable, always-on processing.

Tools for efficient inference. Binarised neural networks [75] reduce weights and activations to one bit; FINN [76] compiles them into streaming FPGA dataflow (up to 12.3 million MNIST classifications per second); integer-only quantisation [77] made 8-bit inference standard; hls4ml [78] translates trained networks into FPGA firmware with latencies around 100 ns for particle-physics triggers. Quantised networks still need integer multipliers (FPGA DSP blocks); our INT8 MLP baseline represents this approach.

Work	Year	Platform	Signals	Hardware (as reported)	Detection (as reported)	Gap for our problem
Verma et al. [60]	2010	ASIC	scalp EEG	9 µJ / feature vector	classifier off-chip	EEG only
Bin Altaf et al. [61]	2015	ASIC	scalp EEG	2.73 µJ / class.	95.7 % sens., epoch-level	patient-specific
Shoaran et al. [64]	2018	ASIC	iEEG	41.2 nJ / class.	F1 87.9 % (block split)	implant, per patient
NeuralTree [65]	2022	ASIC	EEG, iEEG	0.227 µJ / class.	95.6 % sens. (EEG)	neural signals only
Wang et al. [66]	2018	FPGA (Zynq)	scalp EEG	380 mW	98.4 %, 0.356 FD/h	per patient, high power
Farhang Razi & Schmid [68]	2022	FPGA (Cyclone V)	iEEG	0.49 mW dynamic	100 % sens.	per patient, implant
Busia et al. [70]	2025	FPGA (iCE40UP5K)	scalp EEG, 4 ch	4.55 nJ / inference	0.3 FP/h	EEG only, CHB-MIT
e-Glass [71]	2018	MCU	scalp EEG, 4 ch	2.71 days / charge	93.8 % sens. (epoch)	EEG only
EEGformer [72]	2024	MCU (GAP9)	scalp EEG	0.31 mJ / inference	73 %, 0.15 FP/h	multipliers, EEG only
BrainFuseNet [25]	2024	MCU (GAP9)	EEG + PPG + ACC	0.11 mJ / inference	0.21 FP/h	multipliers, no ECG/EMG

Hardware figures are reported in different units (per feature vector, per classification or per inference; dynamic or total power), so they are not directly comparable; we will report energy and LUTs per decision at a fixed decision rate (one per second) together with false alarms per 24 h.

Gap. Almost every hardware detector classifies EEG alone, mostly on paediatric scalp EEG (CHB-MIT) or implanted electrodes, and most are trained per patient and report epoch-level metrics. None maps the whole classifier into FPGA lookup tables, and none is evaluated on multimodal wearable recordings of unseen patients.

2.6 Research gaps and how this project addresses them

#	Gap found in the literature	Evidence	How this project addresses it
1	No LUT-native (weightless, logic-gate or LUT) seizure detector exists	Sections 2.4, 2.5: LUT networks are benchmarked on images, physics and tabular data; the only weightless seizure work is a 4-patient WiSARD study [59]	First benchmark of DWN, LGN, TreeLUT and LogicNets-style networks for seizure detection, on SeizeIT2 (M4)
2	Hardware detectors use EEG alone, although fusion is the most consistent way to cut false alarms	Section 2.2 [8], [22]; Section 2.5	Integer features from behind-the-ear EEG, ECG, EMG and motion in one bit-exact front end (M3)
3	Missing sensors are rarely handled; LUT networks never	Section 2.2 [26]; Section 2.4	Presence bits, modality dropout, reliability-gated and late fusion, and missing-sensor tests (N6)
4	The encoder's hardware cost is ignored, although it dominates small LUT networks	[58]; our estimate: about 850 of 960 LUTs	Cost model that counts every comparator; comparator-budgeted and learned encoders (N2)
5	LUT networks are stateless; seizures unfold over tens of seconds	Section 2.4 [51]	A stateful DWN with state bits fed back every second (N1)
6	Results are patient-specific, leak between training and test, or are scored with incomparable metrics	Section 2.3 [41], [42], [43]	Patient-wise cross-validation, a locked test set used once, SzCORE event metrics with the paper's metrics alongside
7	Small models learn from noisy labels and from larger models only in floating point	Section 2.3 [38], [39]	Noise-robust and multiple-instance losses (N5) and distillation from LightGBM (N4) into a LUT network
8	Detectors are black boxes	Section 2.4	Tracing alarms to the inputs read by each LUT (N7)
9	Published wearable baselines are not independently reproduced	Section 2.2	Independent reproduction of the SeizeIT2 ChronoNet baseline, with and without test-patient

#	Gap found in the literature	Evidence	How this project addresses it
			information (M2)

Chapter 3. Dataset and data audit

3.1 SeizeIT2

We use SeizeIT2 [9], a public multimodal wearable dataset of patients with focal epilepsy, released on OpenNeuro as ds005873 (version 1.1.0, DOI 10.18112/openneuro.ds005873.v1.1.0, licence CC0). The patients were monitored in the epilepsy monitoring units of five European centres (Leuven, Coimbra, Freiburg, Karolinska and Aachen) with the Sensor Dot wearable, while the hospital video-EEG was recorded for the clinical annotation of every seizure. The wearable records four modalities:

Modality	Module (from the channel labels)	Channels	Sampling rate (measured)
EEG	EEG module, electrodes behind the ears (bte-EEG)	2: BTEleft, BTEright or a cross-head channel (CROSStop)	256 Hz
ECG	ECG/EMG module	1	256 Hz
EMG	ECG/EMG module	1	256 Hz
Motion	ECG/EMG module and EEG module	3-axis accelerometer + 3-axis gyroscope per module	25 Hz

The dataset defines the split we follow: **96 training patients (sub-001 to sub-096) and 29 test patients (sub-097 to sub-125)**, balanced by the authors for seizure count, data amount, centre and seizure type. All our model development uses only the 96 training patients, with 5-fold patient-wise cross-validation inside them; the 29 test patients are scored once, at the end (Phase 2).

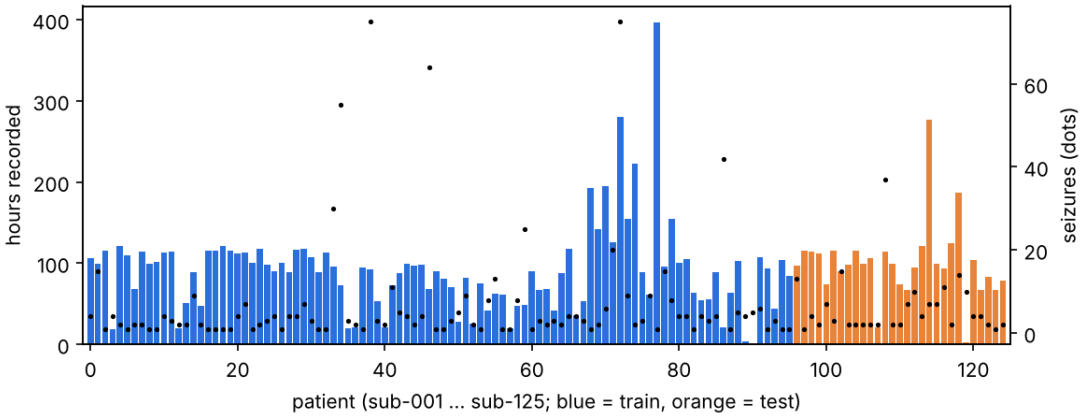
3.2 Why a data audit

Before writing any model code we measured every file of the release (11,009 EDF files, 125.9 GB) instead of trusting the documentation, because a hardware pipeline has to be designed for the exact sampling rates, signal ranges and missing channels. The audit streams each file from the public S3 bucket, measures it and deletes it, so it needs no local copy of the dataset. Every number in this chapter is computed from the files (`reports/data_audit.md`, tables in `results/audit/`).

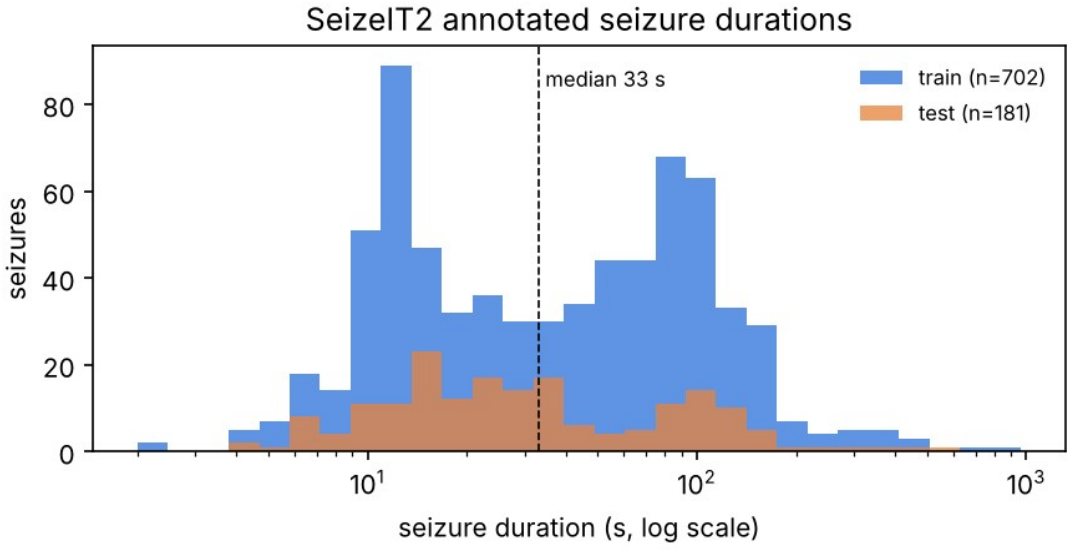
3.3 Inventory

Quantity	Value
Patients	125 (train 96, test 29)
Recordings (runs)	2,850
Files per modality	EEG 2,850, ECG 2,804, EMG 2,804, motion 2,551
Hours of EEG	11,626 h (median 95 h per patient, range 1.1–397 h)
Seizures	883 (train 702, test 181)
Seizure time	14.3 h = **0.123 % of the recording time**
Seizure duration	2 s to 962 s, median 33 s; 87 shorter than 10 s
Missing ECG / EMG	46 runs (3.1 % of the time)
Missing motion	299 runs (3.9 % of the time)

Seizure class	Train	Test	Total
Focal impaired awareness (FIA)	289	102	391
Focal aware (FA)	274	42	316
Focal, awareness unknown	105	10	115
Focal to bilateral tonic-clonic (FBTC)	33	22	55
Unknown onset	1	5	6



Hours of recording and number of seizures per patient



Distribution of seizure durations

Two facts shape the whole project. First, seizures are rare: 0.123 % of the time, so a detector that never fires is 99.88 % "accurate", and accuracy is useless as a metric (Chapter 4.6). Second, most seizures are focal (only 55 of 883 are tonic-clonic), and focal seizures are the hardest to see on two behind-the-ear electrodes.

3.4 Where the files differ from the documentation

Item	Documentation	Files	What we did
Sampling rate of EEG/ECG/EMG	250 Hz	**256 Hz** in every file	All windows are defined in seconds; every integer filter is designed for 256 Hz
Number of seizures	886 (train 704, test 182)	**883** (train 702, test 181)	We report our count and cite the paper's
Seizure classes	includes "subclinical"	no such class in the event files	Breakdowns use the classes present
Shortest seizure	3 s	2 s	Kept
Female patients	51	62 in `participants.tsv`	Reported as a discrepancy
Centre and age per patient	used for stratification	**not in the public release**	Folds stratified by seizure count only
Seizure visibility on the wearable	expected	not annotated (all 883 n/a)	Visible-seizure analysis dropped
Gaps inside files	"discontinuous"	every EDF is continuous	Gaps exist only between runs; a window never crosses a file boundary

3.5 Signal quality

- **Artefact rule of the official baseline.** The baseline discards windows whose RMS is outside 13–150 µV. On behind-the-ear channels the typical RMS is close to the 13 µV floor (median per-run RMS 8.0 µV and 12.1 µV for the two channels), so the rule flags 69 % and 47 % of ordinary windows. We also found that the released evaluation code combines the two channels with a Python `and` on two lists, which returns only the second list, so in practice only one channel is gated. We reproduce this behaviour for the baseline and, for our own models, feed the artefact flags as input bits instead of gating blindly (decision D21).
- **Motion sensors** are exactly constant for 22.8 % of channel time (sensor idle), so a flat motion signal does not mean a missing file; presence bits come from file availability.
- **Truncated downloads.** A later check (Chapter 6) found 9 recordings silently cut short by incomplete S3 reads; all were re-downloaded and every cached recording is now verified against the audit durations.

3.6 Consequences for the design

1. Design for 256 Hz read from the file header, never a hard-coded rate.
2. Treat each modality as possibly missing: every model receives a presence bit per modality.
3. Choose operating points and post-processing for an event rate of a few seizures per 100 hours.
4. Use patient-wise splits only: windows of one patient never appear in both training and validation.

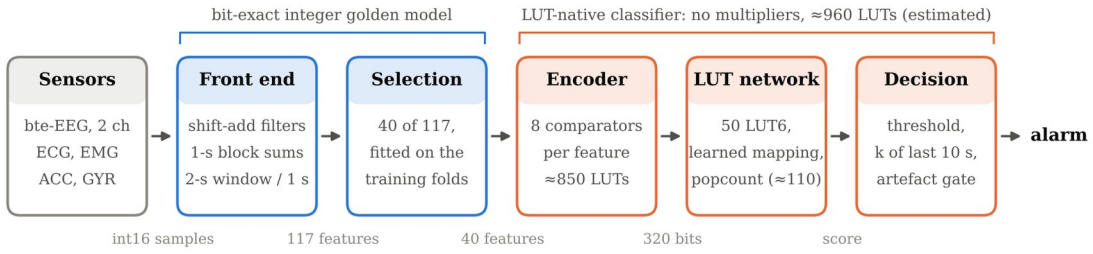
Chapter 4. Proposed system and methodology

4.1 Design goals

The detector is meant to run on a small FPGA inside a wearable, next to the sensors. That fixes five requirements:

1. **Integer arithmetic only**, from the raw samples to the alarm, so the software model and the hardware give bit-identical results.
2. **No multipliers in the classifier.** Multipliers (FPGA DSP blocks) dominate area and power in neural-network accelerators; a lookup-table (LUT) network uses the FPGA's native logic element instead.
3. **A budget of a few thousand LUTs** for the whole chain (features excluded): small enough for a low-power FPGA.
4. **Inter-patient generalisation:** trained once on other patients, it must work on a new patient without recalibration.
5. **Missing sensors are normal:** about 3–4 % of the recording time has no ECG/EMG or motion (Chapter 3).

4.2 System overview



Block diagram of the proposed detector (estimated LUTs of the base configuration)

Every second the front end produces one feature vector describing the last 2 seconds (2-s windows, 1-s hop). The encoder turns each selected feature into 8 bits by comparing it with 8 thresholds; the LUT network maps those bits to an integer score; the decision logic raises an alarm when enough recent seconds are positive.

4.3 Integer feature extraction (golden model)

The front end is written as a **golden model**: a bit-exact software description of the hardware (`src/features/golden.py`, specification in `spec/fixed_point.md`). Inputs are int16 samples (0.25 µV per bit for biopotentials). Filters are second-order sections with coefficients

quantised as sums of powers of two (so a “multiplication” is a few shifts and adds) and error feedback to keep rounding noise below one bit. Windows are built from two 1-s block accumulators, so means are shifts. Logarithms are the position of the leading one plus three mantissa bits.

Group	Count	Features (per 2-s window)
EEG	84	per channel and across channels: mean absolute value, line length, mobility, zero crossings, extrema count, band amplitude in delta, theta, alpha, beta and high-frequency bands; channel sign agreement; artefact flags
ECG	5	log2 of the RR interval (heart rate), log2 of the mean absolute successive RR difference, a signal-quality flag
EMG	8	mean absolute value, line length, zero crossings, high/low frequency ratio
Motion	16	accelerometer and gyroscope magnitude, jerk, activity count, band amplitude
Presence	4	one bit each for ECG, EMG, accelerometer and gyroscope being available

Many features also have a **context** version: the difference between the feature and a causal 64-s exponential moving average of itself, so the network sees “change from this patient’s recent baseline” (for example, heart rate rising). Heart rate is computed from integer R-peak timestamps without any division.

A floating-point twin of the same pipeline is kept to measure what the integer arithmetic costs (Chapter 5).

4.4 The LUT-native classifier: Differentiable Weightless Neural Network

A weightless neural network has no weights to multiply. Each neuron is a truth table: it reads a few input bits, forms an address and outputs the bit stored at that address. On an FPGA a 6-input truth table is exactly one LUT6, the basic logic element, so one neuron costs one LUT.

The Differentiable Weightless Neural Network (DWN) [10] makes such networks trainable by gradient descent:

- **Thermometer encoding.** Feature f becomes bits b(f,i) = [x_f > t(f,i)] for thresholds t(f,1) < … < t(f,8) placed at quantiles of the training data. In hardware each bit is one comparator against a constant.
- **LUT layer with learnable mapping.** Each of the L neurons reads n = 6 bits. Which 6 bits it reads is learned during training; the 64 table entries are learned with the Extended Finite Difference gradient, and a straight-through estimator passes gradients between layers.
- **Popcount score.** The neurons are split into a seizure group and a background group; the score is popcount(seizure group) − popcount(background group), an integer computed with an adder tree.

Our base configuration (decision D33/D34): **L = 50 LUT6 neurons in one layer, 8 thresholds per feature, learnable mapping, the 40 most useful features chosen per training set**. Estimated cost about 960 LUTs in total, of which the thermometer encoder (comparators) is about 850 and the network itself 50: the encoder, not the network, dominates. We re-implemented the DWN in pure PyTorch so it trains on CPUs as well as GPUs; it reproduces the reference CUDA kernel exactly (unit tests).

Feature selection. The 40 inputs are chosen inside each training set only (never using validation patients), first by mutual information and, after the interpretability study, by the gain importance of a LightGBM model trained on the same windows (Chapter 5).

4.5 Models compared

Tier	Models	Role
Published baselines	ChronoNet [30] (deep CNN + GRU on raw EEG), SVM on 42 EEG features [7]	The SeizeIT2 paper's reference results
Tier 1: software references	logistic regression, random forest, XGBoost, LightGBM, CatBoost	Best accuracy achievable on our integer features, ignoring hardware
Tier 1: hardware-oriented	INT8 quantised MLP (uses multipliers), TreeLUT [57], logic-gate network [11], LogicNets-style network [12], DWN [10]	Accuracy per LUT
Tier 1: other references	GRU on features; 1-D CNN and TCN on raw EEG	Sequence and raw-signal references
Tier 2: our extensions to the DWN	N1–N7 (below)	Each kept only if it is significantly better than the base

Tier-2 extensions:

ID	Idea	Question
N1	Stateful LUT network: 16 state bits fed back every second	Does memory of the last minute help?
N2	Encoder design: uniform, Gaussian or learned thresholds under a comparator budget	Can the encoder (the largest cost) be cheaper or better?
N3	Label-free per-patient threshold	Can the detector adapt to a new

ID	Idea	Question
	adaptation	patient without labels?
N4	Distillation from LightGBM and teacher-guided hard negatives	Can the small network learn from the strong one?
N5	Seizure-aware losses: focal, noise-robust (GCE, SCE), multiple-instance learning, onset weighting	Do loss functions for rare, noisily labelled events help?
N6	Reliability-gated fusion, modality dropout, late fusion per modality	Does the detector survive missing sensors?
N7	Interpretability: which inputs do the LUTs read during true and false alarms?	Why does it fail, and what should change?

4.6 Evaluation protocol

- **Split.** The official split (96 training, 29 test patients). All design choices use 5-fold cross-validation over the 96 training patients, split by patient (each fold's validation patients are never seen in its training). The 29 test patients are scored once, at the end, with every choice frozen; any script that touches them needs an explicit flag and logs the access.
- **Event scoring (SzCORE [15]).** A detected event counts as correct if it overlaps a reference seizure, with 30 s of tolerance before onset and 60 s after the end; events closer than 90 s are merged. Reported per patient and averaged: sensitivity (share of seizures detected), precision, **event-F1**, and **false alarms per 24 hours**. Window-level AUROC and AUPR are reported as well.
- **Decision logic.** One decision per second; an alarm when at least k of the last 10 decisions are positive, with an optional artefact gate. The threshold, k and the gate are chosen on the pooled out-of-fold predictions of the training patients, for two operating points: **best event-F1**, and **highest sensitivity with at most one false alarm per hour**.
- **Statistics.** Methods are compared per patient with the paired Wilcoxon signed-rank test and a bootstrap 95 % confidence interval over patients. A Tier-2 method is adopted only if $p < 0.05$ in its favour.
- **Hardware cost.** An analytical estimate per model: comparators of the encoder, LUTs of the network, adders of the popcount and post-processing; multipliers counted separately. Phase 2 replaces estimates with synthesis results.
- **Paper metrics.** For the comparison with the SeizeIT2 paper we also implement its own metrics (any-overlap sensitivity, false alarms per hour, AUROC, AUPR) exactly as in the released code, with two defects of that code corrected in a separate column.

Chapter 5. Work done so far and results

The project is organised in eight milestones (M1–M8). Phase 1 of the B.Tech project covers M1–M4 and most of M5; Phase 2 covers the rest of M5, the final test (M6), the hardware handoff (M7), the FPGA implementation and the paper (M8). All numbers in this chapter come from result files in the repository (`results/`), and all model results are from cross-validation on the 96 training patients: **no model of ours has been scored on the 29 test patients yet**, by design.

5.1 Status by milestone

Milestone	Content	Status
M1	Repository, data loader, full data audit of SeizeIT2	Done
M2	Reproduction of the published baselines	ChronoNet done; SVM waiting for the 42-feature definition of Vandecasteele et al. [7]
M3	Integer (hardware) feature pipeline and its floating-point twin	Done
M4	Comparison of model families on 5-fold cross-validation	Mostly done (9 families); large DWNs, GRU, LogicNets-style, raw-EEG CNN/TCN running
M5	Our extensions N1–N7	Running: N1, N3, N7 done; 17 variants of N1/N2/N4/N5/N6 in the queue
M6	Final test on the 29 held-out patients, 5 seeds	Phase 2 (code ready)
M7	Hardware handoff: export, test vectors, Verilog check	Phase 2 (code ready and tested)
M8	Paper	Phase 2

5.2 Software built

About 8,600 lines of Python (`src/`, `scripts/`) and 1,500 lines of tests (91 test functions), run automatically on every change (continuous integration on synthetic data, since the real data cannot be in the repository):

- **Data layer:** our own EDF reader that keeps the raw integer samples (bit-exact against pyEDFlib), BIDS discovery of recordings and events, the official split, patient-wise folds.
- **Audit tools** that stream all 11,009 files from S3 and measure them without storing the dataset.
- **Evaluation package:** SzCORE event scoring (checked identical to the reference `timescoring` library), the SeizeIT2 paper's metrics ported literally from its code, operating-point selection, paired statistics.

- **Baselines:** a PyTorch port of ChronoNet (129,122 parameters, as the original Keras model) with the official pre-processing.
- **Integer feature pipeline** (golden model) with its fixed-point specification and a floating-point twin.
- **Models:** a pure-PyTorch DWN, logic-gate network, TreeLUT, INT8 MLP, LogicNets-style network, GRU, CNN/TCN, and all the Tier-2 options as configuration switches; one training function shared by cross-validation and the final runs, so the final model is trained exactly as it was evaluated.
- **Experiment management:** every run writes its configuration, git commit, log and metrics; runs resume after an interruption at fold and epoch level; a decision log records every keep/drop decision with its evidence.
- **Hardware handoff tools:** export of a trained DWN as truth tables and thresholds, generation of test vectors for every pipeline stage, a Verilog generator and an Icarus Verilog testbench (100 % agreement with the software on the test fixture).

5.3 Baseline reproduction (M2)

We re-ran the paper's deep-learning baseline, ChronoNet, on the 29 test patients (the only use of the test patients so far, logged). The released code chooses its training epoch using these same test patients; we reproduced that ("official") and also trained a "clean" version that uses the last epoch and no test information.

	Seizures detected (any-overlap)	False alarms / hour	AUROC	AUPR
Published [9], Table 2	84.2 %	100.5	0.792	0.104
Ours, official protocol	74.5 %	22.6	0.741	0.109*
Ours, clean protocol	67.1 %	11.9	0.746	0.111*

* mean over the 90 test recordings that contain a seizure; pooled over all windows the AUPR is 0.004–0.005.

The published numbers are **not fully reproduced**: sensitivity is 10 points lower, while false alarms are four to eight times lower. The AUPR matches when it is averaged per recording, which suggests that this is how the paper computed it; the paper's false-alarm rate matches neither of the two definitions in its code. We report the reproduction as measured, and the clean version is the ChronoNet result our models will be compared with in Phase 2. Scored with SzCORE (the metric we use for our models), clean ChronoNet reaches sensitivity 0.79 at 106 false alarms per day, event-F1 0.027.

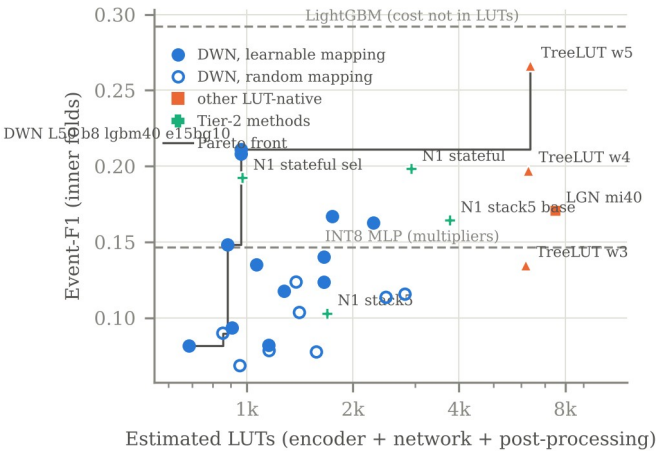
5.4 Integer features cost nothing measurable (M3)

On 10 training patients, the integer features correlate with their floating-point twins at a median of 0.9965, the artefact flags agree on 99.7 % of windows, and a LightGBM classifier trained on either reaches the same leave-patients-out AUROC (0.736 integer vs 0.735 float). The hardware arithmetic loses no accuracy.

5.5 Comparison of model families (M4)

5-fold cross-validation on the 96 training patients; event metrics at each model's best-F1 operating point; estimated hardware cost.

Model	AUROC	Event-F1	Sensitivity	False alarms / 24 h	Sens. at ≤ 1 FA/h	Estimated cost
LightGBM	0.870	**0.292**	0.327	3.6	0.594	~30.6k LUTs (not a hardware design)
CatBoost	0.863	0.286	0.337	3.0	0.618	~27.7k LUTs
XGBoost	0.867	0.277	0.286	3.4	0.603	~47.9k LUTs
TreeLUT (5-bit leaves)	0.857	0.266	0.329	4.2	0.619	~6.4k LUTs
DWN, 50 LUT6 (our base)	0.840	**0.211**	0.315	**2.4**	0.581	**~960 LUTs, 0 multipliers**
Random forest	0.852	0.198	0.216	0.5	0.536	~323k LUTs
Logic-gate network	0.843	0.171	0.330	4.9	0.540	~7.5k LUTs
INT8 MLP	0.830	0.147	0.225	7.8	0.520	9,600 multiplications
Logistic regression	0.806	0.105	0.216	8.8	0.409	117 multiplications



Event-F1 against estimated LUTs

- The DWN is the most compact model by a wide margin: about 6.6 times fewer LUTs than TreeLUT and 8 times fewer than the logic-gate network, with no multipliers, and it has one of the lowest false-alarm rates.
- It is still below the best tree models (LightGBM: per-patient event-F1 difference significant, p = 0.004). Closing this gap is the purpose of M5.
- The INT8 MLP, the usual way to put a neural network on a chip, is worse than the DWN on every metric while needing 9,600 multiplications per window.

5.6 Designing the DWN

- **Size.** A sweep over 50, 100 and 200 LUTs × 4, 8, 16 thresholds × learned or random input mapping (17 of the 18 configurations finished; the last one is queued) showed no significant gain from size between 50 and 200 LUTs; 4 thresholds were worst at every size; learned mapping beat random mapping in 7 of the 8 completed pairs. The smallest equivalent model, 50 LUTs with 8 thresholds, became the base (D33). Sizes 500–2,000 are being trained on the GPU.
- **Feature selection.** The interpretability study (N7, below) showed that mutual-information selection kept about 34 EEG features and missed the heart-rate and muscle features that the tree models rely on. Selecting the 40 inputs by LightGBM importance instead (inside each training set) raised event-F1 from 0.148 to 0.208 ($p = 0.047$) and cut false alarms from 7.0 to 2.3 per day ($p < 10^{-6}$) at the same size (D34).
- **Training recipe.** 15 epochs with 10 background windows per seizure window gave the same results as 30 epochs with 20 ($p = 0.88$) in a quarter of the time, so it is the base recipe for all further experiments.

5.7 First results of our extensions (M5)

Extension	Result	Decision
N7 interpretability	During true and false alarms the network reads almost the same inputs: false alarms look like seizures to it. Its input usage agrees moderately with LightGBM's SHAP importance (Spearman 0.41). It revealed the feature-selection gap above.	Led to D34
N3 per-patient adaptation	Re-fitting the thresholds on each new patient's first 1, 2 or 4 hours	Not adopted

Extension	Result	Decision
	raised false alarms in every variant (e.g. 6.8 → 10.2–39.5 per day at 1 h)	
N1 stateful network (16 state bits)	Event-F1 0.192 vs 0.211 ($p = 0.66$); false alarms 1.1 vs 2.4 per day ($p = 0.013$); AUROC 0.861 vs 0.840	Not adopted alone (F1 not better); considered for the combination
N2, N4, N5, N6	17 variants in the queue	Pending

Where the base detector works and where it fails (out-of-fold, best-F1 operating point):

Seizure type	Detected
Focal to bilateral tonic-clonic	29 of 33 (88 %)
Focal impaired awareness	50 of 289 (17 %)
Focal aware	26 of 274 (9 %)
Focal, awareness unknown	8 of 101 (8 %)

Sensors removed at test time	Event-F1 (base 0.211)
no ECG	0.092 ($p = 2 \cdot 10^{-5}$)
no motion	0.136 ($p = 0.004$)
no EMG	0.199 (n.s.)
EEG only	0.074 ($p = 2 \cdot 10^{-6}$)

The detector catches most tonic-clonic seizures, the dangerous ones, but few focal seizures, and it depends heavily on the heart signal and motion. Robustness to missing sensors (N6) is therefore a central question of M5.

Chapter 6. Challenges faced and how we overcame them

Every problem below is recorded with its evidence in the project's decision log (`results/decisions.md`, decision numbers D1–D35 in brackets). Failed ideas are kept there as well.

6.1 Data

Challenge	What happened	How we overcame it
The data did not match its documentation	The paper and our project brief say 250 Hz and 886 seizures; the files say 256 Hz and 883. Centre and age labels, needed for stratified splits, are not in the public release; seizure-visibility labels are empty.	A full audit of all 11,009 files before any modelling (Chapter 3). Every window and filter is defined in seconds and designed for 256 Hz; folds are stratified by seizure count; analyses the data cannot support were dropped and reported (D3, D4, D6).
126 GB of data, a cloud machine with neither the disk nor a GPU	The dataset could not be stored where the code runs.	The audit and the feature extraction stream each file from the public S3 bucket, process it and delete it; only the 3.5 GB of integer features are kept.
Silent download truncation	Comparing integer and float features exposed one ECG file cut at 51,855 of 71,835 s. A new verifier then found 9 recordings silently truncated by incomplete network reads (one EEG file had 12 of 16,363 windows).	Downloads are checked against the expected size, the EDF header is checked against the file size, and every cached recording is verified against the audit durations. All 9 were re-extracted; the 3 model runs trained on the bad cache were marked invalid and re-run (D30).
A recording with an ECG lead but no detectable heartbeat	The feature extractor crashed on an empty beat list (3,250 s of data).	The recording is kept, its ECG features are set to 0 and a quality bit is raised (D28).
The official artefact rule does not suit behind-the-ear EEG	The 13 µV RMS floor flags 69 % of ordinary windows on one channel, because bte-EEG amplitudes are close to it.	For our models the artefact flags are input bits, and whether to gate alarms on them is chosen on the training folds (D21).

6.2 Reproducing the published baseline

Challenge	What happened	How we overcame it
ChronoNet did not reproduce	Our run detected 74.5 % of seizures against 84.2 % published, with 22.6 instead of 100.5 false alarms per hour.	We ported the official code literally (verified on 900 random cases), then checked the metric definitions: the published AUPR matches only when averaged per recording (0.109 vs 0.104), and the published false-

Challenge	What happened	How we overcame it
		alarm rate matches neither definition in the code. The gap is reported as measured, not tuned away.
The released code selects its model on the test patients	Its training loop picks the epoch with the best score on the same patients it is then evaluated on.	We reproduced it as published and also trained a “clean” version (last epoch, no test information), which is the honest reference for our final comparison (D13).
Defects in the released evaluation code	True positives can be counted more than once, and the artefact gate (`a and b` on two Python lists) silently uses only one channel.	Reproduced as written for the comparison with the paper; corrected numbers reported in a separate column (D8, D11).
The SVM baseline cannot be rebuilt yet	The dataset paper defers its 42 features to Vandecasteele et al. [7], whose full text could not be retrieved from the coding environment.	Still open: we need that paper’s feature table (D15).

6.3 Hardware arithmetic

Challenge	What happened	How we overcame it
Rounding noise in integer filters	With plain rounding, the 0.5 Hz high-pass filter had an error of 140 LSB standard deviation, larger than the signal (61 LSB).	Second-order error feedback of the discarded bits: error 0.7 LSB (D18).
Quantised coefficients moved the filter zeros	Rounding each coefficient independently moved the low-pass zeros away from Nyquist (stopband floor −55 dB).	Structure-preserving quantisation, b = b0·(1, ±2, 1), with coefficients as sums of powers of two (D19).
Heart rate without a divider	Heart rate needs 60/RR.	We use log2(RR) from integer R-peak timestamps: no division, no multiplication (D22).
Proving the integer pipeline costs nothing	Hardware arithmetic could lose accuracy silently.	A floating-point twin of every feature: median correlation 0.9965 and equal LightGBM AUROC (0.736 vs 0.735).
TreeLUT scores were not comparable across folds	Each fold’s integer score had its own scale, so pooled thresholds were meaningless; 3-bit leaves could not represent the model.	Scores converted to a common unit; leaf width tuned on the folds (5 bits) (D29).
Verilog disagreed with the software	Thresholds outside a feature’s bit width (e.g. 79 on a 2-bit presence feature) produced wrong comparator logic.	The generator now emits constant bits for such thresholds; the simulation agrees with the software on 100 % of test vectors.

6.4 Compute and infrastructure

Challenge	What happened	How we overcame it
No GPU in the cloud; a small GPU laptop	The cloud machine has 4 CPU cores; the laptop has a 6 GB GPU and 7.4 GB of RAM under WSL. The reference DWN library only runs on CUDA.	We re-implemented the DWN in pure PyTorch, identical to the library's CUDA kernel (unit-tested), so it runs on both. CPU jobs (small models, trees) run in the cloud; large networks and raw-EEG models run on the laptop GPU (D24, D27, D32).
Long runs interrupted	Container restarts lost the logic-gate network run after 3 of 5 folds; others were cut mid-fold.	Every run resumes at fold and epoch level; a queue skips finished runs and continues unfinished ones; a resumed run was checked to give identical metrics.
Out-of-memory kills	The stateful network's sequence sampler held 24 million windows in memory (13.9 GB of a 15 GB machine).	A new sampler reads each needed recording once and stores bits as bytes: same sequences, peak memory 3.1 GB.
Two machines, one code base	The laptop and the cloud must not overwrite each other's work.	One git branch; the laptop only sends finished results; written instructions for the laptop side; automatic tests on every change.

6.5 Machine learning

Challenge	What happened	How we overcame it
Seizures are 0.123 % of the time	Accuracy is meaningless; training on all windows teaches the model to never fire.	Event-based scoring (SzCORE) per patient; training on all seizure windows plus a fixed ratio of background windows; operating points chosen for a false-alarm budget.
Results must hold on unseen patients	Mixing a patient's windows between training and validation inflates results.	Patient-wise 5-fold cross-validation; feature selection and thresholds fitted inside each training fold only; the test patients are locked behind a flag and every access is logged (D14, D25, D26).
The network ignored the heart signal	The interpretability study showed that mutual-information selection kept about 34 EEG features and missed the heart-rate and muscle features.	Feature selection by LightGBM importance: event-F1 0.148 → 0.208, false alarms 7.0 → 2.3 per day (D34).
A confounded comparison	The stateful network had silently received all 117 features instead of the selected 40, so its apparent gain mixed two changes.	Found by checking the run's metadata. All models now share one training function; the run was repeated; the correction is logged next to the original claim.
Stacking past windows picked	Ranking 585 stacked columns kept	Rank on the current window, then

Challenge	What happened	How we overcame it
duplicate features	lagged copies of only 11–13 distinct features.	keep every lag of the winners (D31).

6.6 Still open

- **Focal seizures** are mostly missed (8–17 % detected at the best-F1 point), while tonic-clonic seizures are caught (88 %). The M5 extensions and the larger networks address this.
- **The gap to LightGBM** (event-F1 0.211 vs 0.292) is not closed yet.
- **Robustness to missing sensors:** without ECG, event-F1 falls from 0.211 to 0.092 (N6 addresses this).
- **SVM baseline** and **centre labels** depend on information outside the public release.

Chapter 7. Plan for Phase 2

Phase 2 turns the cross-validated design into final, test-set results and a working hardware implementation. The 29 test patients stay untouched until every design choice is frozen, so the final numbers are an honest estimate of performance on new patients.

7.1 Work packages

#	Work package	Content	Output
1	Finish the extensions (M5)	The 17 queued variants of N1, N2, N4, N5 and N6; missing-sensor tests for the fusion variants; a combination run of the extensions that win; large DWNs (500–2,000 LUTs); GRU, LogicNets-style and raw-EEG CNN/TCN references; the SVM baseline if its features become available	Ablation table with per-patient statistics; final choice of the DWN design
2	Final test (M6)	Retrain each final model (best of every family, the chosen DWN, clean ChronoNet) on all 96 training patients with 5 random seeds; score each once on the 29 test patients	Test-set table (SzCORE and paper metrics), seizure-type and missing-sensor breakdowns, confidence intervals
3	Hardware handoff (M7)	Export the final DWN (thresholds, input mapping, truth tables, popcount head); test vectors for every stage from raw samples to alarms; reference Verilog checked against the software	Handoff package with 100 % bit agreement
4	FPGA implementation	Synthesis and place-and-route on a low-power FPGA (device chosen with the guide); measured LUTs, flip-flops, clock frequency, latency and estimated power; the INT8 MLP and TreeLUT baselines on the same device for comparison	Hardware results table; demonstration of a recording streamed through the design

#	Work package	Content	Output
5	Paper and final report (M8)	Conference paper (target: EMBC 2027; full papers reportedly due 24 January 2027, to be confirmed on the official site) and the final B.Tech report	Paper draft, final report, presentation

7.2 Timeline

Period	Work
Weeks 1–2 after the Phase 1 review	Work package 1: remaining M5 experiments and decisions
Week 3	Work packages 2 and 3: final test and hardware handoff
Weeks 4–8	Work package 4: FPGA implementation and measurements
Weeks 8–10	Work package 5: paper, final report, demonstration

The machine-learning experiments run unattended on the cloud machine and the laptop GPU, so most of work package 1 proceeds in parallel with the Phase 1 review.

7.3 Expected outcome

A seizure detector for multimodal wearable signals that

- uses integer arithmetic end to end and **no multipliers in the classifier**;
- fits in **about one thousand LUTs** for the classifier and encoder (current estimate: about 960);
- is evaluated **on 29 unseen patients** with event-based metrics, against the published ChronoNet baseline, gradient-boosted trees and an INT8 neural network;
- states exactly which seizure types it detects and how it behaves when a sensor is missing.

7.4 Risks

Risk	Mitigation
The DWN stays below the tree models	Report it honestly as an accuracy-per-LUT trade-off (the DWN is about 6.6 times smaller than TreeLUT); the extensions and larger networks are designed to narrow the gap
Focal seizures remain mostly undetected	Report sensitivity per seizure type, and the same breakdown for the published ChronoNet baseline on the same test patients
Test results lower than cross-validation	Expected to some degree; the test set is used once and reported as measured
FPGA results differ from the estimates	The estimates are replaced by synthesis results; the handoff package and test vectors make debugging deterministic

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